Mastiyage Don Sudeera Hasaranga Gunathilaka

sudeera.gunathilaka@aist.go.jp Website Google Scholar
Languages: English (Fluent), Japanese (Fluent), Sinhala (Native)
Reviewer: IEEE

RESEARCH INTERESTS

Unconventional computing architectures (coherent, quantum, quantum-inspired etc.)
Artificial intelligence (LLM for scientific discovery, numerical optimisation, etc.)
High-performance computing (GPU computing, large-scale simulation, etc.)

EDUCATION

Ph.D., Artificial Intelligence, Tokyo Institute of Technology, Japan — Mar 2025
M.Eng., Artificial Intelligence, Tokyo Institute of Technology, Japan — Mar 2022
B.Eng., Information Science, Shonan Institute of Technology, Japan — Mar 2020
H.S.Dip., Physical Science, Nalanda College, Colombo 10, Sri Lanka — Aug 2015

WORK EXPERIENCE

Research Scientist (Tenured), National Institute of Advanced Industrial Science and Technology – G-QuAT — Apr 2025 – Present

RESEARCH EXPERIENCE

Research Intern, NTT PHI Laboratories, CA, USA — Jun 2021 - Mar 2025
Advisors: Dr. Yoshitaka Inui, Prof. Timothee Leleu, Prof. Yoshihisa Yamamoto
Term 1: Jun 2021 – Aug 2021 and Term 2: Jul 2023 – Mar 2025

- Incorporated Zeeman terms into Gaussian SDE models of coherent Ising machines
- Improved solution success probability on benchmark optimisation problems
- Developed GPU-parallelised CUDA code for large-scale CIM simulations
- Experimented on Wishart Planted Ensembles for Ising machines

Intern, Jij Inc., Tokyo, Japan — Feb 2022 – Mar 2022
Advisors: Dr. Yu Yamashiro,

- Evaluated Hybrid Simulated Quantum Annealing performance
- Benchmarked optimisation accuracy and convergence behaviour

Research Assistant, Tokyo Institute of Technology, Japan — Nov 2020 – Dec 2021
Advisors: Prof. Toru Aonishi,

- Improved compressed sensing image reconstruction accuracy with CIMs
- Worked on quantum–classical hybrid system (CIM–CDP)
- Proposed the closed-loop CIM models for with effective Zeeman term realization

PUBLICATIONS

[1] Aonishi, T., Nagasawa, T., Koizumi, T., **Gunathilaka, M.D.S.H.**, Mimura, K., Okada, M., Kako, S., Yamamoto, Y. *Highly versatile FPGA-implemented cyber coherent Ising machine.* IEEE Access. LINK 2024

[2] **Gunathilaka, M.D.S.H.**, Inui, Y., Kako, S., Yamamoto, Y., Aonishi, T. *Mean-field coherent Ising machines with artificial Zeeman terms.* Journal of Applied Physics, 134(23):234901. LINK 2023

[3] **Gunathilaka, M.D.S.H.**, Kako, S., Inui, Y., et al. *Effective implementation of ℓ_0-regularised compressed sensing with chaotic-amplitude-controlled coherent Ising machines.* Scientific Reports 13, 16140. LINK 2023

[4] Inui, Y., **Gunathilaka, M.D.S.H.**, Kako, S., et al. *Control of amplitude homogeneity in coherent Ising machines with artificial Zeeman terms.* Communications Physics 5, 154. LINK 2022

[5] **Gunathilaka, M.D.S.H.**, Mahboubi, S., Ninomiya, H. *Acceleration technique of two-phase quasi-Newton method with momentum for optimization problems.* ThinkMind. LINK 2020

PREPRINTS

[1] Leleu, T., **Gunathilaka, M.D.S.H.**, Ghimenti, F. and Ganguli, S., *Contrastive Concept-Tree Search for LLM-Assisted Algorithm Discovery.*. arXiv:2602.03132. LINK 2026

[2] **Gunathilaka, M.D.S.H.**, Inui, Y., Kako, S., Mimura, K., Okada, M., Yamamoto, Y. and Aonishi, T., *L0-regularized compressed sensing with Mean-field Coherent Ising Machines.*. arXiv:2405.00366. LINK 2024

TALKS AND POSTER PRESENTATIONS

Talk, International Workshop on Ising Machines 2026, Singapore — May 2026
Invited Talk, Atsumi foundation workshop 2025, Nirasaki — Jul 2025
Talk, International Workshop on Ising Machines 2025, Chicago — May 2025
Talk, International Network on Quantum Annealing 2024, Tokyo — Oct 2024
Talk, Asia Future Conference, Bangkok (*Best Presentation Award*) — Aug 2024
Talk and Poster, Adiabatic Quantum Computing 2024, Glasgow — Apr 2024
Talk, NTT Retreat Meeting, San Francisco — Apr 2024
Invited Talk, NTT–RIKEN Workshop on Photonics, Neural Networks, etc. — Feb 2024
2 Talks, Spring Meeting of the Physical Society of Japan 2024, Online — Mar 2024
Talk, 2nd Quantum Annealing workshop, Tokyo — Dec 2023
Poster, Quantum Information Processing 2024, Taipei — Jan 2024
Poster, StatPhys28, University of Tokyo, Japan — Aug 2023
Invited Talk, NTT Basic Research Laboratories, Atsugi — Jun 2023
Poster, Conference on Neural Computation, Stanford University — Oct 2022

TEACHING EXPERIENCE

Student Assistant, Shonan Institute of Technology — Apr 2019 – Mar 2020
- Assisted teaching a course in artificial intelligence
- Assisted teaching a course in logic circuits and Java programming

HONORS, SCHOLARSHIPS AND AWARDS

Outstanding Student Award — 2019
Atsumi International Foundation Scholarship — 3,000,000 JPY
Kobayashi Scholarship Foundation Scholarship — 6,480,000 JPY
Rotary–Yoneyama Memorial Scholarship (Club Support) — 1,680,000 JPY
Rotary–Yoneyama Memorial Scholarship — 1,200,000 JPY
Monbukagakusho Honors Scholarship — 576,000 JPY

EXTRA ACTIVITIES (2025 ONWARDS)

16th QS conference volunteer — Oct 2025
CEATEC AIST booth volunteer — Oct 2025
Supercomputing conference 2025 mentor — Nov 2025
Member of the Japanese delegation to the UN International Year of Quantum event in the UK — Nov 2025